

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			A chloroplast B nucleus NOT nucleolus	1			1		1
	(b)	(i)		Any four (x1) from: A. (The cytoplasm of) the amoeba has a {lower/ more negative} water potential than the {surrounding water / external environment}/ ORA (1) B. Water will therefore move into the cell by <u>osmosis</u> . (1) If not credited here, can be given with reference to vacuole C. (By pumping ions into the vacuole) {it lowers the water potential of the vacuole / creates a water potential gradient between the cytoplasm and the vacuole} (1) D. Water will move from the {cytoplasm/ cell} to the vacuole (by osmosis) (1) E. (The water is then expelled from the cell) which prevents {osmotic lysis/ bursting} (1) Reject reference to turgidity		2	2	4		
		(ii)		Mitochondria (1) To provide the ATP for {active transport <u>of ions</u> / pumping <u>of ions</u> / exocytosis}(1) Ribosomes(1) Produce proteins for {transport of ions/ carrier proteins}(1)		4		4		
	(c)			(The cell wall is) {strong / resists tension/ generates pressure potential/ owtte} and prevents {osmotic lysis/ bursting} (1) - Beta glucose (1) - It is made of (long) straight chains / alternate molecules {inverted/ rotated 180° }(1) {hydrogen bonds / cross-links} between (parallel) chains/ to form microfibrils (1)	2	2		4		